TEXORA · SESSIONAL NOTES · LAB EXPERIMENT SHEET

Presentation and Visualization — Graphs, Charts, 2D, 3D

Choosing the chart that tells the truth about a set of numbers, building it in the spreadsheet, and presenting the result on slides — including when a glossy 3D chart is a lie in disguise.

CSE-2106 · FUNDAMENTALS OF COMPUTING SESSIONAL · EXPERIMENT 2 · LEVEL 2 TERM 1

1. Aim

To turn tabulated data into correct, readable charts (column, line, pie, scatter) in MS Office/Open Office, to assemble a short presentation around those charts, and to judge when 2D must be preferred over 3D.

2. Concepts you need

CHART	USE IT FOR	DO NOT USE IT FOR
Column / bar	Comparing values across categories (production by machine, marks by subject)	Showing a trend over many time points — use a line
Line	Trend over time (monthly production, temperature during a process)	Parts of a whole
Pie	Shares of one whole that sum to 100% (fibre mix of a mill's consumption)	Comparing similar-sized values; more than ~6 slices; values that do not sum to a whole
Scatter (XY)	Relationship between two measured quantities (yarn count vs strength)	Categories on an axis — scatter needs numbers on both axes

- **Parts of a chart:** title (states the message, not just the topic), axis titles with units, data series, legend (only when more than one series), data labels, and the source note.
- **2D vs 3D:** a 3D effect tilts the plotting area, so front bars look bigger and pie slices at the front look larger than equal slices at the back. 3D adds no information and subtracts accuracy — professional

reports stay in 2D. Use 3D only where depth itself is the data (a true 3D surface), never as decoration.

- **Axis honesty:** a column chart's value axis starts at zero; a truncated axis exaggerates differences.

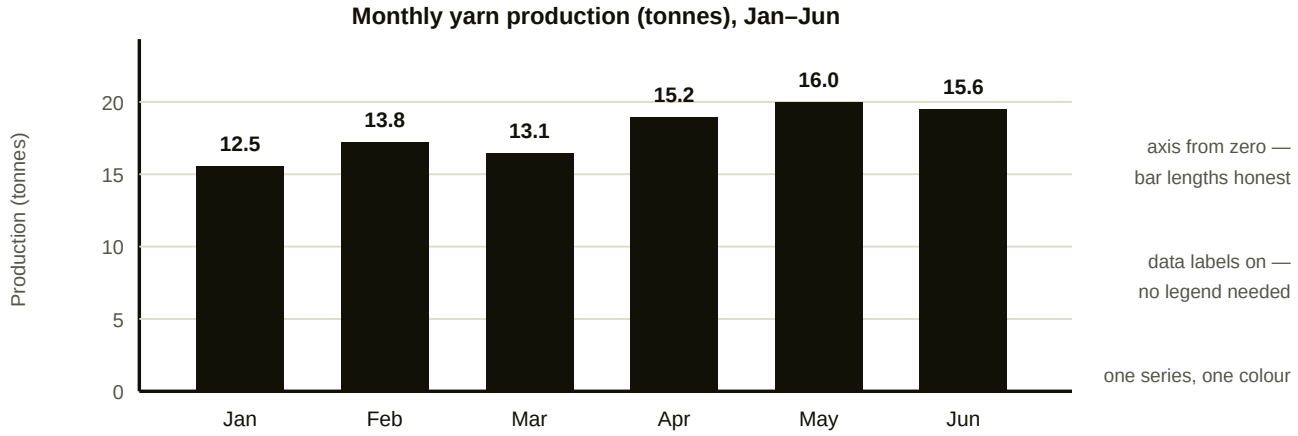


Figure 1 — The column chart of the worked example, drawn to the rules of Section 2. A line chart of the same six points would be equally correct — the data is a time series — but the columns make the month-to-month comparison the report needs.

3. Lab procedure

1. Enter the worked-example data (month, production in tonnes) into a worksheet.
2. Select the data including headings; insert a **column chart**. Add the chart title, the axis title with units, and data labels. Confirm the value axis starts at 0.
3. Change the chart type to **line**; note what changes in the message (comparison → trend). Change back to column.
4. Enter the fibre-consumption data (Cotton 60%, Polyester 25%, Jute 10%, Others 5%) and insert a **pie chart** with percentage labels. Check the slices sum to 100% before charting.
5. Rebuild the pie as a **3D pie**. Rotate it and observe: the slice facing the viewer looks larger than an equal slice behind. Record this as your evidence for the 2D rule, then delete the 3D version.
6. Start a presentation (PowerPoint/Impress): build five slides — title; agenda; the method in three bullets; the column chart slide (paste the chart; title states the finding); conclusion slide with the pie chart. Apply one consistent design template, body text no smaller than 18 pt, slide titles no smaller than 28 pt.
7. Run the slide show; check every slide is readable from the back of the room and no text overflows its box.
8. Save the presentation and export it to PDF.

4. Worked example — the data and its checks

MONTH	JAN	FEB	MAR	APR	MAY	JUN	TOTAL
Production (tonnes)	12.5	13.8	13.1	15.2	16.0	15.6	**86.2**

CHECKS BEFORE ANYTHING IS PRESENTED

Total = 12.5 + 13.8 + 13.1 + 15.2 + 16.0 + 15.6 = **86.2 tonnes**; six-month average = 86.2 ÷ 6 ≈ **14.4 tonnes/month**. Pie data: 60 + 25 + 10 + 5 = **100%** — a pie of anything else would be an error, not a chart. The conclusion slide's claim must be one the chart proves: "Production rose from 12.5 t in January to a peak of 16.0 t in May, easing to 15.6 t in June" — every number in that sentence is readable off Figure 1.

5. Practice tasks

1. For each data set, choose the correct chart and justify the choice in one line: (a) marks of one student in six subjects; (b) a dye-house's water temperature recorded every 10 minutes for 2 hours; (c) share of a factory's energy cost borne by four departments; (d) fabric strength measured at 12 different yarn counts.
2. A colleague's column chart shows two machines producing 48 and 52 kg, but the value axis starts at 40, making the second bar look three times taller. State what is wrong and the two honest ways to present the comparison.
3. Write the exact title and the one-sentence conclusion for a slide presenting the pie chart of fibre consumption.

Answer key

1. (a) **Column** — comparing values across categories (subjects). (b) **Line** — a quantity changing continuously with time. (c) **Pie** — shares of one whole (100% of energy cost), four slices. (d) **Scatter** — both variables are measurements; the chart shows whether strength depends on count.
2. The axis is truncated, so bar lengths are no longer proportional to values — the reader's eye compares lengths, not labels. Honest presentations: start the value axis at zero, or switch to a form that does not encode value by length from a baseline (e.g. a simple table or dot plot) while stating the actual figures 48 and 52.
3. Title: "Fibre consumption by type — share of total (%)". Conclusion: "Cotton is the mill's dominant fibre at 60% of consumption, with polyester second at 25%, so fibre-price risk is concentrated in cotton." (Any conclusion is acceptable if every figure in it appears in the chart.)